

Derivatives Quiz

Calculus Version A

Name: _____

1. (4 pts) Let $f(x) = x^4 - 8x^2 + 7$. Find $f'(x)$.

Answer: $f'(x) = 4x^3 - 16x$

2. (10 pts) Let $f(x) = x^4 - 8x^2 + 7$. Answer each question below.

a. Find $f'(x)$.

$4x^3 - 16x$

b. Find $f''(x)$.

$4(3x^2 - 4)$

c. Find all critical points of f .

$x = -2, 0, 2$

d. Classify each critical point as a local min, local max, or neither.

3. (6 pts) Given $x^2 + y^2 = 25$, use implicit differentiation to find $\frac{dy}{dx}$.

Answer: $\frac{dy}{dx} = -\frac{x}{y}$

4. (8 pts) Given $y = x^{y^x}$, use implicit differentiation to find $\frac{dy}{dx}$.

Answer: $\frac{dy}{dx} = -\frac{x^{y^x-1}y^{x+1}(x\log(x)\log(y)+1)}{x^{y^x+1}y^x\log(x)-y}$